

Chapter 1

Lecture 1

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1.1 Introduction – 1/13

$p_\theta(x_i) = \theta^{x_i}(1 - \theta)^{1-x_i}$ piecewise trick.
 $p_\theta = \frac{1}{\binom{n}{t}}$ not dependent on x

1.2 Introduction - 1/15

Model $x \propto f(x)$ distribution unknown, parameterized by unknown θ . By looking at x , make inference θ . Dependency of x on θ is only on the distribution.

Statistic: $T = T(x)$ function of random variable; does not depend on θ .

T $f_\theta(t)$ potentially not as accurate as the inference made from x . When mapping T is invertible, there is no difference between x and t .

If information is sacrificed (T loses information on θ).

Sufficient statistic;

In case where mapping does not lose any information, that statistic is called a ‘sufficient’ statistic. Invertible mapping preserves all data.

Want sufficient statistic that compresses data and preserves information...

Sufficient: If conditional distribution does not depend on θ , $f_\theta(x|t)$ $p_\theta(x|t) = \frac{p_\theta(x)}{p_\theta(t)}$

Fisher-Neyman factorization theorem: allows us to identify sufficient statistics directly from the likelihood function $f_\theta(x)$. can be taken as a working definition of sufficiency THM: let $f_\theta(x)$ be a pdf or pmf of X . Stat $T = T(X)$ is sufficient for θ iff $\exists g_\theta(t)$ and $h(x)$ s.t.: $f_\theta(x) = g_\theta(T(x))h(x)$ where $h(x)$ does NOT depend on θ

Bernoulli-trials revisited: PMF of X $p_\theta(x) = \theta^{T(x)}(1 - \theta)^{n-T(x)}$ here x only shows up as part of $T(x)$ therefore by the Fisher-Neyman thm, $T = \sum_{i=1}^n X_i$ is a sufficient statistic for θ .

X is discrete, so

$$p_\theta = \sum_{y: T(y)=t} p_\theta(y)$$

thus for any (x, t) such that $T(x) = t$, we have

$$p_\theta(x|t) = \frac{h(x)}{\sum_{y: T(y)=t} h(y)}$$

now ratio does not depend on θ , so by definition $T = T(x)$ is a sufficient statistic for θ .

If $p_\theta(x) = h(x)g_\theta(T(x))\dots$

Example: Gaussian with unknown mean. Suppose we observe $X = [X_1, \dots, X_n]^t$, where

$$X_i \text{ iid } \mathcal{N}(\mu, \sigma^2)$$

where σ^2 is a known constant and μ unknown parameter

We have $f_\mu(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$ and hence,

$$f_\mu = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(\frac{\mu}{\dots}\right) = g_\mu(T(x))h(x)$$

Example: both μ and σ^2 unknown and $\theta = [\mu, \sigma^2]^t$ is a two-dimensional vector

We have $f_\theta(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp(\dots)$

with $T(x) = [\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2]$

Example: Uniform with unknown parameter

$$X_i \text{ iid } \mathcal{U}(0, \theta)$$

with $\theta > 0$ unknown parameter, want to make inference on upper-bound

We have

$$f_\theta(x_i) = \frac{1}{\theta} 1_{x_i \leq \theta}$$

for any $x_i \geq 0$, hence

$$f_\theta(x) = \frac{1}{\theta^n} \prod_{i=1}^n 1_{x_i \leq \theta} = \frac{1}{\theta^n} 1_{\max_{i \in [n]} x_i \leq \theta}$$

Therefore $T(X) = \max_{i \in [n]} X_i$ is a sufficient statistic

A sufficient statistic is minimal if it is a function of any other sufficient statistics

1.2.1 Certification of minimality

The minimality of a sufficient statistic can be verified based on the following proposition:
a sufficient statistic T is minimal if for any $x, y \in \mathcal{X}$,

$$\frac{f_\theta(x)}{f_\theta(y)} \text{ is independent of } \theta \implies T(x) = T(y)$$

if the ratio of their likelihoods is independent of θ , then need to check if it implies $T(x) = T(y)$. Only a sufficient condition.

Note that

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} = \left(\frac{\theta}{1-\theta} \right)^{T(x)-T(y)}$$

Clearly, the ratio does not depend on θ iff $T(x) = T(y)$.

1.3 Lecture week 2 – Jan 20

Any function of the observation that preserves all the information is a sufficient statistic.

Minimal sufficient statistic

A sufficient statistic is *minimal* if it is a function of any other sufficient statistics. This is the "optimal" data compression.

A sufficient condition for minimality given sufficiency: The minimality of a sufficient statistic can be verified based on the following proposition: A sufficient statistic T is minimal for any $x, y \in \mathcal{X}$,

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} \text{ is independent of } \theta \implies T(x) = T(y)$$

(Proof) Let T be a sufficient statistic that satisfies the above property. To show that T is minimal, we must show that for any other statistic T' , T is a function of T' .

Now suppose that $T'(x) = T'(y)$. We have

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} = \frac{g'_{\theta}(T'(x))h'(x)}{g'_{\theta}(T'(y))h'(y)} = \frac{h'(x)}{h'(y)}$$

which is independent of θ

Previously, we assume T is a sufficient statistic and then provides a sufficient condition for the minimality.

Let $T = T(X)$ be a stat. It can be shown that T is a sufficient statistic if for any $x, y \in \mathcal{X}$,

$$T(x) = T(y) \implies \frac{f_{\theta}(x)}{f_{\theta}(y)} \text{ is independent of } \theta.$$

Proof based on Fisher-Neyman...

Combined with the previous proposition, a statistic $T = T(X)$ is a minimal sufficient statistic if for any $x, y \in \mathcal{X}$,

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} \text{ is independent of } \theta \iff T(x) = T(y)$$

Example 1.3.1 (Bernoulli trials). Consider the statistic

$$T = \sum_{i=1}^n X_i$$

Note that

$$\frac{f_{\theta}(x)}{f_{\theta}(y)} = \left(\frac{\theta}{1-\theta} \right)^{T(x)-T(y)}$$

Clearly, the ratio $\frac{f_{\theta}(x)}{f_{\theta}(y)}$ does not depend on θ iff $T(x) = T(y)$.

Complete sufficient statistic

A statistic $T = T(X)$ is complete if for any real-valued function ϕ ,

$$\mathbb{E}_{\theta}[\phi(T)] = 0, \forall \theta \in \Theta \implies \mathbb{P}_{\theta}(\phi(T) = 0) = 0, \forall \theta \in \Theta$$

Example 1.3.2 (Bernoulli trials). Recall

$$T = \sum_{i=1}^n X_i$$

is a sufficient statistic for θ .

If for some real-valued function ϕ we have

$$\mathbb{E}_{\theta}[\phi(T)] = \sum_{t=0}^n \phi(t) \binom{n}{t} \theta^t (1-\theta)^{n-t} = 0$$

we must have

$$\sum_{t=0}^n \phi(t) \binom{n}{t} \left(\frac{\theta}{1-\theta} \right)^t = 0$$

for all $\theta \in (0, 1)$.

Example 1.3.3 (Uniform with unknown support). Recall that for uniform random variables over an unknown support $[0, \theta]$.

$$T = \max_{i \in [n]} X_i$$

is a sufficient statistic for θ . The pdf of T can be calculated as

$$f_{\theta}(t) = nt^{n-1}\theta^{-n}1_{0 < t \leq \theta}$$

If for some real-valued function ϕ we have

$$\mathbb{E}_\theta[\phi(T)] = n\theta^{-n} \int_0^\theta \phi(t)t^{n-1}dt = 0$$

for all $\theta > 0$, we must have

$$g(\theta) = \int_0^\theta \phi(t)t^{n-1}dt = 0, \quad \forall \theta > 0$$

Taking the derivative, $g'(\theta) = \phi(\theta)\theta^{n-1}$.